Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Intelligent CNC Turning Automation Using AI-Based Process Expertise			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical·Materials			
	Medium Category	Small Category	Medium Category	Small Category
	Precision manufacturing machinery	Cutting Machine Tools, Numerical Control (NC) Units, CAD/CAM-related Software		
Concept of Proposed Technology	The proposed solution is an AI-supported software system tailored to precision turning, aimed at enabling reliable and explainable automation of CNC process planning and program creation. Instead of requiring detailed NC programming expertise, the system operates on a minimal and application-focused input set provided by the user. This input includes the description of the part geometry (via parametric data, CAD models, or 2D technical drawings), the material condition of the workpiece (material type and hardness), boundary conditions related to available cutting tools and tool holders, and the required quality targets such as dimensional accuracy and surface finish. Incoming geometric and quality information is first processed by a manufacturing-oriented interpretation layer, which transforms heterogeneous input data into a structured internal representation suitable for turning process planning. This representation explicitly encodes machining-relevant elements such as functional diameters, axial and radial transitions, grooves, tolerance regions, and surface quality constraints. For 2D drawings, AI-based semantic analysis techniques are applied to identify and interpret geometric entities, dimensional annotations, and tolerance specifications, enabling automated and consistent extraction of turning-relevant features [1]. Building on this structured representation, a knowledge-driven process planning module applies validated turning expertise to derive feasible machining solutions. This module determines appropriate turning strategies, tool and insert configurations,			

	admissible cutting parameter ranges, and path-related constraints, while explicitly respecting quality and safety requirements. The result is a controller-independent process plan characterized by deterministic rules, explicit constraints, and fully traceable decision logic, allowing transparent verification of all planning steps [2]. The transition from process planning to executable CNC code is handled by a constrained large language model (LLM) that functions exclusively as a controlled translation mechanism. Based on predefined syntax templates, controller-specific rules, and parameter boundaries, the LLM converts the validated process plan into CNC programs for industrial controllers such as Siemens SINUMERIK. Automated validation routines ensure syntactic correctness, logical completeness, and full compatibility with the target controller before program release [3]. To further enhance robustness and industrial usability, the system incorporates process diagnostics and risk evaluation functions. These functions assess the planned operating conditions with respect to known turning-related failure mechanisms, including dynamic instabilities, progressive tool wear, surface integrity issues, and dimensional deviations. If potential risks are identified, the system suggests permissible adjustments within the validated process domain. Final responsibility remains with the operator: a human-in-the-loop concept requires explicit approval based on an explainable summary of the selected strategies, constraints, and identified risks prior to CNC program execution.
Rationale for Joint R&D Project	Industrial manufacturing is increasingly constrained by a systemic lack of skilled CNC programming personnel, a problem that is especially pronounced in precision turning. Programming turning operations extends far beyond the correct use of NC syntax: it requires integrated understanding of cutting physics, tool–workpiece interactions, insert and tool-holder selection, heat generation, wear progression, and mechanisms governing dimensional accuracy and surface integrity [4]. Such competence is typically acquired through long-term experience and remains largely tacit and operator-specific. As a result, it is difficult to replicate, transfer, or scale—posing a particular challenge for small and medium-sized enterprises (SMEs) operating multiple products, machines, or sites. Research activities at the proposing institute have shown that formalized expert knowledge and structured decision frameworks can convert experience-based turning know-how into explicit rules, admissible operating ranges, and quality-oriented constraints. These methods make it possible to decouple process reliability from individual experts, while maintaining robust machining behavior and consistent quality outcomes [2], [5]. Despite this progress, an unresolved bottleneck persists at the interface between validated turning process plans and their execution on industrial CNC systems. Existing approaches to automated NC code creation rarely incorporate comprehensive turning-specific process intelligence, enforceable safety limits, or explainable decision paths. Consequently, their use in high-precision and quality-sensitive turning applications remains limited [6]. In industrial practice, the final step—translating an expert-approved process plan into controller-specific CNC code—still depends heavily on manual programming by highly experienced specialists. This step represents a critical barrier to productivity, scalability, and wider industrial adoption. The proposed project addresses this gap by introducing a strictly constrained, AI-supported large language model (LLM) as a technical translation layer between validated process planning and CNC execution. The LLM is deliberately restricted to a code-generation role and does not perform autonomous process decisions. Instead, it converts expert-system-approved turning plans into controller-specific CNC programs (e.g., Siemens SINUMERIK) using predefined templates, syntax rules, parameter boundaries, and safety constraints. This controlled use of AI enables efficient and repeatable CNC program creation while ensuring determinism, traceability, and operator supervision through a human-in-the-loop concept. Successful industrial deployment of such an approach requires more than algorithmic development. It demands reliable software architectures, seamless CNC controller connectivity, secure data management, and intuitive user interfaces that fit established shop-floor workflows. These requirements align closely with the core competencies of Korean SMEs and startups, which have strong capabilities in CNC systems, industrial software platforms, digital manufacturing solutions, and scalable product development. Joint R&D is therefore indispensable. By combining the proposing institute's process-centered expertise in turning and expert-system design with the Korean partners' strengths in industrial integration and platform development, the project enables the realization of a practically deployable, safe, and explainable solution for AI-assisted CNC program generation in precision turning. This collaborative approach directly supports TIPA's mission by accelerating the transfer of advanced manufacturing technologies into SME production, lowering adoption risks, and delivering a market-ready solution that responds to workforce shortages, productivity pressures, and the demand for consistent quality in modern manufacturing.

(max 10 pages)

Key Words	1. CNC Turning Process 2. Knowledge-Based Expert and Assistant Systems 3. AI-Assisted and Constrained Artificial Intelligence 4. Large Language Model (LLM) 5. Human-in-the-Loop Manufacturing Systems

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	• To design and implement a flexible, knowledge-driven expert system for precision turning that can be configured for a wide range of turning operations and machine tool setups, including but not limited to external and internal turning, facing, and grooving. • To enable the automated determination of suitable cutting tools, insert geometries, machining strategies, and admissible cutting parameter ranges (cutting speed, feed, depth of cut) based on part geometry, workpiece material condition, and specified quality targets. • To develop AI-supported methods for extracting and interpreting manufacturing-relevant features from CAD data and 2D technical drawings, including semantic understanding of geometric entities and dimensional specifications required for turning-oriented process planning. • To incorporate systematic diagnostics and corrective logic addressing typical turning-related process issues such as chatter, built-up edge formation, surface quality deviations, dimensional errors, and instability caused by tool wear. • To realize controller-aware, LLM-supported CNC program generation, operating under strict syntactic, semantic, and safety constraints, ensuring deterministic behavior, verifiable output, and transparent decision chains. • To integrate the developed functionalities into a secure, scalable, and operator-focused software environment that can be connected to industrial CNC turning systems and deployed in real production settings.			
R&D Target Specifications	• Demonstrated reduction of CNC programming effort exceeding 70% relative to conventional manual programming approaches for representative precision turning use cases. • Substantial decrease in scrap and rework risk (target >80%) achieved through knowledge-based selection of cutting tools, machining strategies, and validated operating ranges for cutting speed, feed, and depth of cut, in combination with deterministic CNC code generation. • Consistent fulfillment of defined quality requirements, including dimensional accuracy and surface finish targets (e.g., surface roughness Ra ≤ 0.8 µm in finishing operations), verified through industrial trials and pilot production tests. • Compatibility with industrial CNC control systems, with initial implementation for Siemens SINUMERIK and a modular software design that allows straightforward extension to additional controller types and machine configurations. • Deterministic and transparent CNC program generation, including constrained LLM-based code translation, ensuring syntactic correctness, controller compliance, and complete traceability of process decisions, parameter choices, and applied safety constraints.			
R&D Plan	The proposed R&D activities are organized within a three-year project framework, allowing sufficient time for structured technology development, industrial system integration, and comprehensive validation of the AI-assisted CNC program generation approach for precision grinding. The detailed project timeline, sequencing of tasks, and definition of milestones will be jointly developed and finalized with the Korean industry partner prior to project initiation, taking into account the partner's technical priorities, existing infrastructure, and planned commercialization pathway. All described work packages, task definitions, proposed leadership assignments, and technical specifications are intended as a conceptual baseline. These elements will be closely aligned with the Korean development partner, refined where necessary, and further elaborated collaboratively throughout the project to ensure an optimal technical realization and strong industrial relevance. The overall methodology is based on the integration of four closely interconnected components: i. Domain expertise in turning technology, implemented through a combination of rule-based and data-driven expert systems to capture validated process knowledge; ii. AI-assisted interpretation of geometric and quality-related features derived from CAD models and 2D technical drawings, enabling automatic identification of manufacturing-relevant information; iii. Constrained, AI-assisted CNC program generation using large language models (LLMs) operating within strictly defined syntactic, semantic, and safety boundaries to ensure reliable and verifiable CNC code output; and iv. Industrial-grade CNC system integration, including standardized controller interfaces, validation and approval			

workflows, and operator-oriented user interfaces designed for shop-floor deployment.

Safe automation with human-in-the-loop validation is the core principle of the project. Turning process knowledge is first formalized into deterministic expert rules, quality-driven constraints, and validated parameter windows. These machine-independent process plans constitute the sole input to CNC code generation. LLMs are used exclusively as controlled translation tools to generate controller-specific CNC programs (e.g., Siemens SINUMERIK) based on predefined templates, cutting parameter limits, and controller logic. Autonomous decision-making by the LLM is explicitly excluded. All generated CNC programs undergo deterministic validation and require explicit operator approval prior to execution.
The system is designed to be configurable and extensible, enabling adaptation to different turning operations (e.g., facing, longitudinal turning, contour turning, grooving), machine configurations, and CNC controllers.

Work Packages, Tasks, and Methodologies:
WP1 – Expert System Development for Turning Process Planning
Lead (Optional-Suggestion): Research Institute
Objectives:
Develop a validated knowledge-based expert system for deterministic turning process planning based on part geometry, material properties, and quality requirements.
Key Tasks:

- Identification and classification of representative turning operations (e.g., roughing, finishing, profiling, grooving)
- Systematic formalization of turning process knowledge, including:
  a. Cutting tool and insert selection (material, coating, geometry)
  b. Tool holder and clamping constraints
  c. Cutting strategy selection
  d. Validated cutting parameter windows (cutting speed, feed rate, depth of cut)
- Definition of quality-driven constraints (surface roughness, dimensional tolerances, stability limits)
- Encoding of turning knowledge using rule-based and hybrid (rule + data) expert system approaches

Methodology:

- Expert interviews and structured knowledge acquisition
- Literature-based consolidation of turning process models
- Rule-based expert system design with traceable decision logic
- Validation using reference turning experiments and historical process dataalidation using reference experiments and historical process data

Milestone M1:
Validated expert system for tool, strategy, and parameter selection.

WP2 – Manufacturing Feature Extraction and Interpretation
Lead (Optional-Suggestion): Joint (Research Institute + Korean Partner)
Objectives:
Develop a robust feature interpretation module that converts CAD data and 2D technical drawings into a structured, manufacturing-relevant representation for turning-specific process planning.
Key Tasks:

- Definition of turning-relevant geometric and functional features (e.g., diameters, shoulders, grooves, contours)
- Interpretation of dimensional tolerances and surface roughness specifications
- AI-supported semantic interpretation of geometric entities and dimensional information in technical drawings
- Mapping of extracted features to turning strategies, tool constraints, and expert-system inputs
- Development of standardized interfaces for CAD model and drawing uploads

Milestone M2:
Validated feature interpretation module providing structured, turning-specific inputs to the expert system.

WP3 – LLM-Based CNC Program Generation for Turning
Lead (Optional-Suggestion): Research Institute
Objectives:
Translate expert-system-validated turning process plans into safe, controller-specific CNC programs.
Key Tasks:

- Definition of structured interfaces between expert system outputs and the LLM input layer.
- Development of controller-specific CNC templates and syntax rules (initial focus: Siemens SINUMERIK).
- Implementation of constrained LLM prompting for:
  a) Turning cycles
  b) Tool calls and offset handling
  c) Cutting parameter assignment

- Integration of deterministic syntax checks, rule compliance validation, and controller compatibility verification.

Milestone M3:
Automatic generation of syntactically correct, controller-compatible, and verifiable CNC turning programs.

WP4 – Process Risk Assessment and Defect Prevention
Lead (Optional-Suggestion): Research Institute
Objectives:
Integrate explainable troubleshooting and process risk assessment capabilities into the automated turning process planning workflow.
Key Tasks:

- Identification and classification of common machining defects (e.g., chatter, form and diamonsional deviations).
- Modeling of cause–effect relationships.
- Development of rule-based diagnostic and decision logic.

Milestone M4:
Integrated troubleshooting module providing explainable risk assessment and corrective actions..

WP5 – Industrial System Integration and CNC Connectivity
Lead (Optional-Suggestion): Korean SME / Startup
Objectives:
Deploy the developed expert system and AI-assisted CNC generation workflow on an industrial-grade software platform connected to CNC turning machines.
Key Tasks:

- Design and implementation of system architecture and operator-oriented user interfaces
- Integration with CNC controllers via industrial communication standards (e.g., OPC UA)
- Implementation of data security, access control, and system reliability mechanisms
- Visualization of generated CNC programs, decision logic, and troubleshooting results

Milestone M5:
Fully integrated prototype system connected to an industrial turning machine.

WP6 – Industrial Validation and Pilot Demonstration
Lead (Optional-Suggestion): Joint
Objectives:
Validate system performance, robustness, and usability under real industrial turning conditions.
Key Tasks:

- Pilot testing on industrial CNC turning machines
- Quantitative evaluation of CNC programming time reduction
- Operator feedback and iterative refinement

Milestone M6:
TRL 7 system validated in an industrial turning environment.

Work Packages Overview (36 Months):

- WP1 – Expert System Development for Turning Process Planning
- WP2 – Manufacturing Feature Extraction and Interpretation
- WP3 – LLM-Based CNC Program Generation for Turning
- WP4 – TProcess Risk Assessment and Defect Prevention
- WP5 – Industrial System Integration and CNC Connectivity
- WP6 – Industrial Validation and Pilot Demonstration

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will assume responsibility for the methodological leadership and scientific integrity of the project, providing the foundational process intelligence required for reliable AI-assisted automation in CNC turning. Its primary function is to ensure that all automated planning and code-generation functions are rooted in validated turning science, transparent reasoning, and industrial safety requirements, rather than empirical heuristics or opaque AI behavior. The institute will transform established academic and applied expertise in turning into a structured, explainable digital knowledge base that can be safely deployed in industrial environments. To achieve this, the research institute will develop a comprehensive digital representation of turning know-how, capturing machining strategies, tool and insert usage, interaction limits between tool and workpiece, and admissible operating ranges for cutting parameters. This representation will be complemented by process-oriented models that relate part geometry, material condition, tool geometry, and cutting parameters to achievable quality outcomes, tool life, and process stability under realistic production conditions. A key responsibility lies in defining quality- and safety-relevant operating boundaries for turning processes. These include permissible surface roughness levels, dimensional tolerance ranges, force and thermal load limits, and stability regions that prevent chatter and other dynamic instabilities. Based on these boundaries, the institute will design a hybrid decision framework in which deterministic expert logic forms the backbone of all process decisions, while AI-based automation is applied only within clearly defined and traceable limits. The research institute will further specify the controlled use of large language models by defining strict interfaces, controller-aware syntax constraints, validation rules, and verification procedures. In this architecture, the LLM is deliberately confined to a supportive translation role, converting validated process plans into CNC code without influencing core process decisions or safety-relevant parameters. In addition, the institute will develop a structured diagnostic and risk-assessment layer that connects typical turning-related defects—such as chatter, accelerated tool wear, built-up edge formation, surface integrity degradation, or form errors—to their underlying causes and permissible corrective measures. This logic will enable systematic troubleshooting within predefined and validated process envelopes. Scientific evaluation will be carried out through targeted laboratory investigations, reference turning experiments, and robustness studies to ensure that the developed methods remain reproducible and transferable across different machines, materials, and application scenarios. The research institute will also actively participate in pilot deployments, supporting data interpretation, model refinement, and knowledge adaptation based on industrial feedback. Finally, the institute will contribute to the long-term impact and sustainability of the project by disseminating results through scientific publications, intellectual property generation, and structured technology transfer documentation, thereby supporting future standardization efforts and continued international research–industry collaboration.
Expected R&R of Korean SME Partner	The Korean SME or startup partner will take primary responsibility for converting the research results into an **industry-ready solution** and positioning the technology for practical use and market adoption. This role focuses on translating conceptual and experimental developments into a **stable, scalable, and commercially viable system** that can be reliably deployed within real SME manufacturing operations. Core responsibilities of the Korean partner include the establishment of a **production-oriented software environment** that supports automated turning process planning and CNC program generation under shop-floor conditions. This includes the development of robust connectivity to industrial CNC systems, with initial implementation for Siemens SINUMERIK controls and a forward-looking, modular design that enables adaptation to other controllers and machine tool platforms. The partner will also be responsible for embedding the various functional modules—such as process knowledge models, geometry and feature interpretation, AI-assisted code generation, and diagnostic support—into machine tool ecosystems in a manner that ensures deterministic behavior, real-time responsiveness, and long-term operational reliability. Particular emphasis will be placed on the design of **intuitive operator interfaces** that clearly communicate system decisions, machining strategies, constraints, and identified risks, thereby supporting acceptance and safe use in daily production. In addition, the Korean SME or startup will implement appropriate mechanisms for **data handling, cybersecurity, system stability, and access control**, ensuring compliance with industrial IT/OT requirements and the practical constraints of SME environments. Industrial pilot operation will be conducted at selected production sites, during which system performance, usability, and robustness will be systematically evaluated. Based on insights gained from pilot deployment, the partner will lead iterative system optimization and contribute to the definition of **post-project deployment pathways**, including product refinement, customization strategies, and market introduction within the Korean manufacturing sector and beyond.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project's deliverables include a validated, industrially deployable system for safe, transparent, and explainable automatic CNC program generation in precision turning, as well as real-production-condition demonstrations. The solution being developed will enable a paradigm shift from manual, experience-driven CNC programming towards knowledge-based, AI-assisted and human-supervised process planning. This will significantly enhance the efficiency, robustness and scalability of turning operations. Technical and Industrial Outcomes • A software solution that can be used in different situations and that can reduce CNC programming time by more than 70%. It can also make sure that the results are always the same, that you can follow the steps taken and that it works with industrial CNC controllers. • Substantial reduction in scrap rates and rework through systematic selection of turning strategies, cutting tool configurations, and validated cutting parameter windows, leading to improved process stability and product quality. • Enhanced process consistency and reproducibility across machines, operators, and production sites, particularly in high-precision turning applications with strict dimensional tolerance and surface integrity requirements. • The incorporation of troubleshooting functions that can be explained, as well as process risk assessment functions, enables defects such as chatter, built-up edge, tool wear-induced surface degradation, thermal effects and form deviations to be detected and mitigated early on. Impact on Korean SMEs and Commercialization Potential • Korean SMEs and startups will gain access to an industrial-grade digital manufacturing platform that can be customized and scaled across different turning machines, customers, and application domains. • SMEs can expand their production capacity and product variety without having to increase their number of highly skilled labourers, thanks to their reduced dependence on scarce expert CNC programmers. • The solution enables commercial differentiation by integrating sophisticated turning process intelligence directly into software products, as opposed to relying on machine hardware upgrades or the expertise of individual operators.. • A strong basis for future extensions to additional machining processes, CNC controllers and digital manufacturing services is provided by the modular platform architecture. Workforce and Societal Impact • The shortage of highly skilled CNC programmers is being tackled by the system, which transforms tacit, experience-based turning knowledge into a digital, reusable, and explainable knowledge base. • Operators remain in control and responsible for final decisions in human-in-the-loop workflows, while benefiting from advanced AI-based decision support. • The strategy is instrumental in facilitating the upskilling of the workforce and the preservation of knowledge over the long term, thereby helping to mitigate the repercussions of demographic change and workforce turnover in the manufacturing sector. Strategic and Long-Term Impact • The project promotes international collaboration between research and industry in Germany and Korea, accelerating the adoption of advanced manufacturing technologies by SMEs. • The development of AI-driven smart manufacturing systems that prioritise safety, explainability and industrial acceptance is supported by the project. • The technology developed establishes a foundation for the future standardisation, certification and cross-sector adoption of AI-assisted CNC programming solutions in precision turning and related machining processes.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization